

JNTU Hyderabad

REGULAR EXAMINATION ? 2026

Compiler Design ? Regular Examination, 2026

Subject Code: CS602 Semester: 6 Regulation: R20

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Lexical Analysis)

1. Compare and contrast the key concepts involved in Lexical Analysis. [12 Marks]
2. Define Lexical Analysis. Explain its significance in Compiler Design with a suitable example. [8 Marks]
3. Describe the working principle of Lexical Analysis with necessary equations. [12 Marks]

SECTION B (Unit 2: Syntax Analysis)

4. Write short notes on Syntax Analysis. [12 Marks]
5. Solve a problem related to Syntax Analysis. Show all the steps clearly. [12 Marks]
6. Derive the expression related to Syntax Analysis and explain each term. [8 Marks]

SECTION C (Unit 3: Semantic Analysis)

7. Compare and contrast the key concepts involved in Semantic Analysis. [12 Marks]
8. Define Semantic Analysis. Explain its significance in Compiler Design with a suitable example. [12 Marks]
9. Define Semantic Analysis. Explain its significance in Compiler Design with a suitable example. [8 Marks]

SECTION D (Unit 4: Intermediate Code Generation)

10. Discuss the engineering applications of Intermediate Code Generation. [8 Marks]
11. Derive the expression related to Intermediate Code Generation and explain each term. [12 Marks]
12. Derive the expression related to Intermediate Code Generation and explain each term. [12 Marks]

SECTION E (Unit 5: Code Optimization and Runtime)

13. Explain Code Optimization and Runtime with a neat diagram/sketch and a suitable example. [12 Marks]
14. Derive the expression related to Code Optimization and Runtime and explain each term. [12 Marks]
15. Discuss the engineering applications of Code Optimization and Runtime. [12 Marks]

***** End of Question Paper *****